Reproducible Experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Identify the run. Keep all outcomes. Say what was reproduced.

Distinguish complete procedure replay,saved-output arithmetic and unreproducible aggregate-only reports.

Keep one habit: Declared seeds → complete versioned scoring code → pictured variation → peer-card swap → independent new-run summary.

MAIN LESSON · SECTION 1

What would another learner need to repeat this result?

> A seed names one part of a procedure,not its complete evidence.

Learning goals

I can record enough detail to reproduce a run.

I can distinguish **reproducibility** from validity.

I can report all planned **seed**s instead of selecting the best.

Recall metric and experiment records

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.
- What must accompany a reported validation score?

Readiness for the versioned scoring replay

- Why should expected test results be written before running the function?

> **My readiness explanation**

End goal

By the end, I can explain reproducible experiments using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A result is reproducible only when another builder can recreate the data, split, code, settings, **seed**, and metric calculation.



LEARNING TARGET Two teams use the same **random seed** but different label files. Their results differ. List the records needed to reproduce the experiment.

Builder name and experiment date:

Readiness for the connected explanation

Readiness for Versioned experiments

This week's end goal

MAIN LESSON · SECTION 2

Save identifiable inputs,procedure and environment

> Preserve failed and less favorable declared runs alongside the best one.

Seed,version and environment explanation

A **seed** can help repeat a random split or initialization, but it is not a magic quality stamp. Save **data version**, split membership, code **version**, settings, pre**process**ing, **seed** and metric definition.

Some software operations remain nondeterministic, so document tolerances and environment too.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines experimental records and **seed**s.

Complete versioned policy-replay context

A fictional activity scoring experiment compares two weight settings on the same development records. Save code **version**, **input version**, weights, tie rule and **output**s.

A label such as v2 is useful only when it refers to an identifiable saved state.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use reproducible experiments to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Apply the connected idea: Versioned experiments

MAIN LESSON · SECTION 3

Distinguish repeating a calculation from recreating a training run

> Make one complete small replay concrete and identify what the aggregate-only record lacks.

Complete declared-run summary

Declared run label	Supplied correct / fixed validation count	Accuracy
seed 11	14 /20	70%
seed 22	16 /20	80%
seed 33	12 /20	60%
Complete summary	All three declared runs retained	Mean70%; range 60–80%, width 20 percentage points

A team declares **seed**s 11, 22, 33 before running. Validation accuracies are 14/20, 16/20, 12/20:70, 80, 60 percent. Mean accuracy is 70 percent, with a 20-percentage-point range.

Reporting only **seed** 22 as the model result hides the declared experiment.

These three classifier results are supplied fictional records. A seed alone cannot recreate the unknown classifier, data or environment; the complete model-training procedure is not supplied. Replaying saved-output arithmetic differs from reproducing model training.

The separate complete A/Bweighted-score program below can actually be rerun with its fixed inputs and two weight versions; it has no random step, so its seed field is not applicable. A changed preference score is not an accuracy improvement.

Complete rerunnable versioned scoring program

```
items = [{"name": "A", "fun": 4, "learning": 1},
         {"name": "B", "fun": 2, "learning": 4}]
for version, fun_weight, learning_weight in [("v1", 2, 1), ("v2", 1, 2)]:
    scores = []
    for item in items:
        scores.append(fun_weight * item["fun"]
                      + learning_weight * item["learning"])
    print(version, scores)
```

Trace v1/v2scores and changed policy

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
v1 weights 2 and 1	A 9; B 8	A ranks first
v2 weights 1 and 2	A 6; B 10	B ranks first

Read each row from left to right and connect the evidence to the stated rule.

The ranking changes because the preference weights changed. That is a policy comparison, not proof that v2 is more accurate.

Keep the same data and constraints when comparing weights, and record why a policy better fits the stated goal.

For the worked records, v1 gives A a score of 2 times 4 plus 1, or 9, and B a score of 2 times 2 plus 4, or 8. v2 gives A 4 plus 2 times 1, or 6, and B 2 plus 2 times 4, or 10. The winner changes because the weights changed.

Keep the records fixed so this explanation can be checked. A different preference policy is not automatically a more accurate model.

Explain the evidence record

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

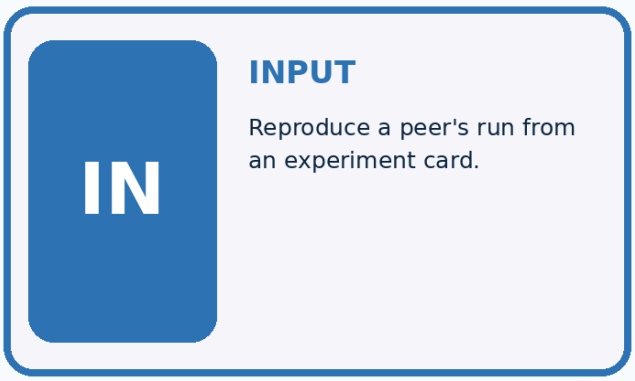
Keep the distinction between an observed result and a proposed next step.

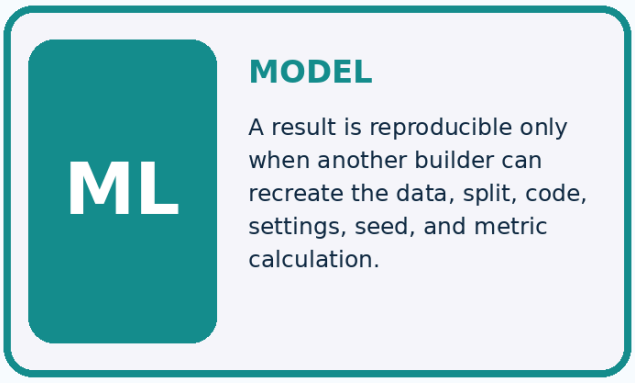
▼ Earlier wording, alternatives and picture transcripts

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Reproduce a peer's run from an experiment card.

MODEL IDEA A result is reproducible only when another builder can recreate the data, split, code, settings, **seed**, and metric calculation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Record data=v2, split=grouped-s1, code=commit A7, **seed**=42. Record pre**process**ing, features, model type, settings, and library **version**s.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Apply the connected idea: Versioned experiments

Compare one planned change at a time

Connect the picture and the explanation

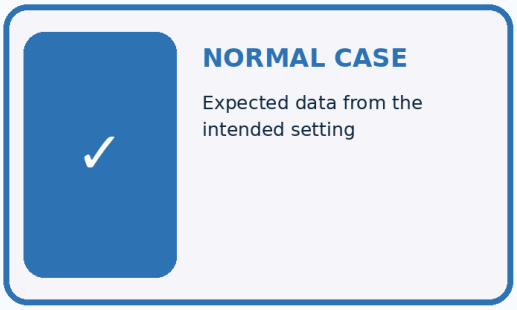
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

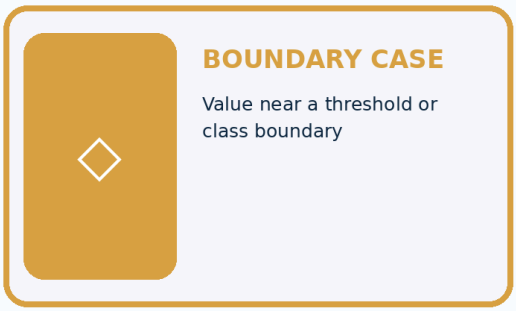
Compare an identifiable experiment with a best-score-only report

> Matching a score alone does not prove matching inputs,settings or predictions.

Four evaluation conditions



NORMAL Expected data and expected behavior.

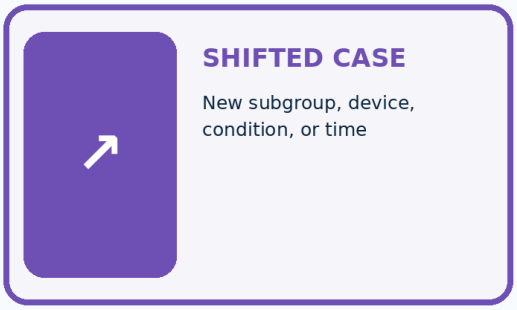


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: The report deletes the **60%** run.

▼ **Check after predicting**

Prediction: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.

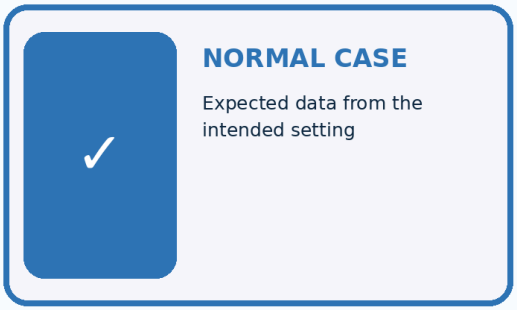
Reason: Removing an unfavourable declared run changes the **evidence** rather than improving the method.

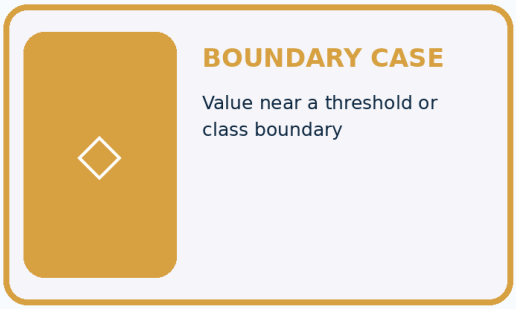
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

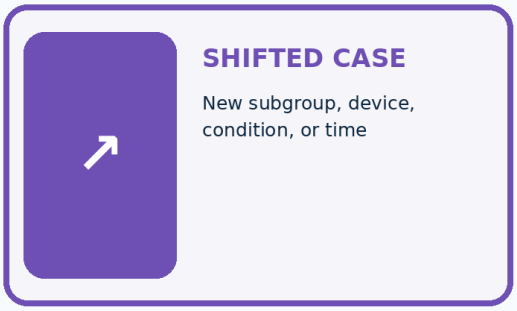
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Keep each run identity,denominator and result beside the summary

> A report should show the variation it summarizes.

Complete pictured runs and separate-fixture scope

Record variation across declared **seed**s instead of selecting the prettiest run.

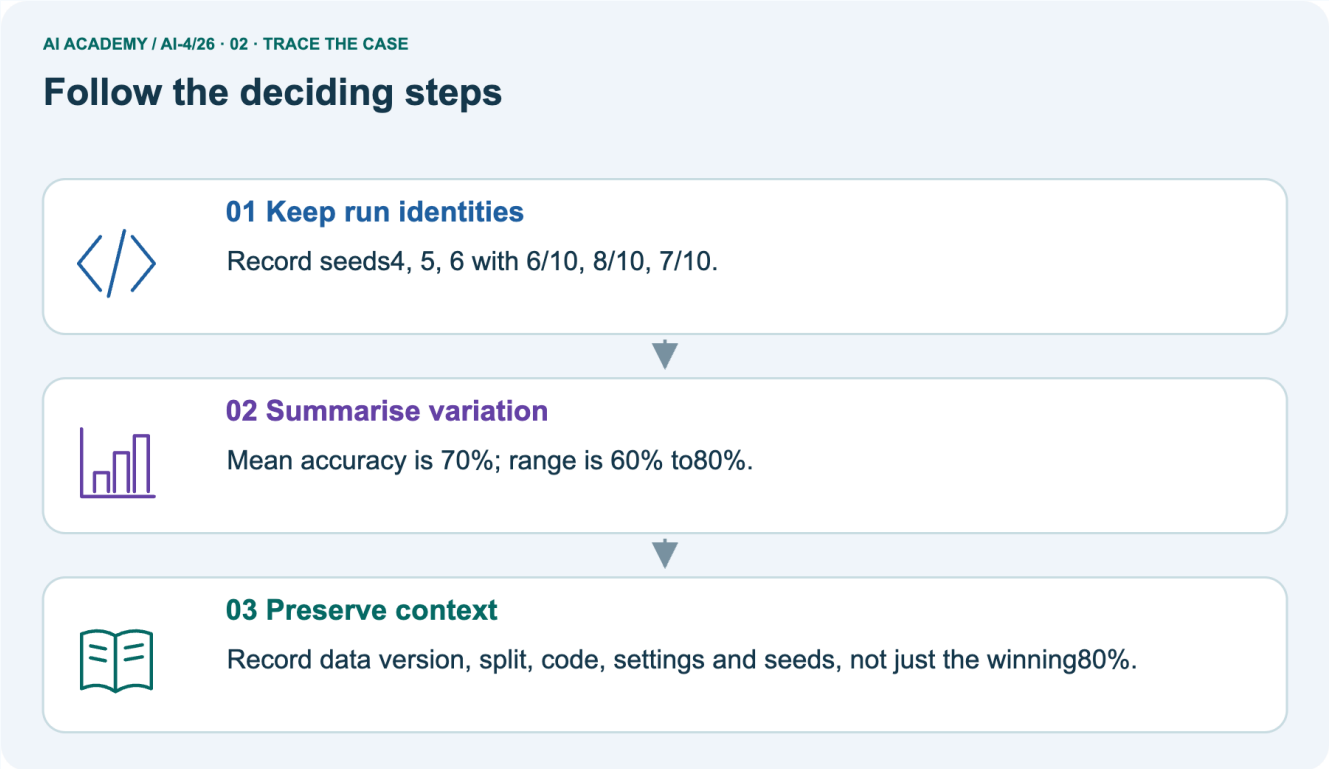
Same fixed validation set of10 cases; declared runs **seed** 4=6 correct, **seed** 5=8, **seed** 6=7. All three are reported.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main supplied runs use seeds 11/22/33 with14/16/12 correct of20 each; mean 70%, range 20 percentage points. The separate picture uses seeds 4/5/6 with6/8/7 correct of10 each, also mean 70%and a60–80%range. Deleting the pictured 60%run leaves 80%and 70%, whose mean 75%is a selectively incomplete summary, not an improved method.

Same denominators make the mean of run accuracies equal pooled correct/total here; with unequal denominators those summaries need not agree. Neither these seeds nor a summary uniquely identify a training procedure.

Trace all results,mean and range



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep run identities	Record seeds4, 5, 6 with 6/10, 8/10, 7/10.
2. Summarise variation	Mean accuracy is 70%; range is 60% to80%.
3. Preserve context	Record data version, split, code, settings and seeds, not just the winning80%.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the original declared-seed graphic

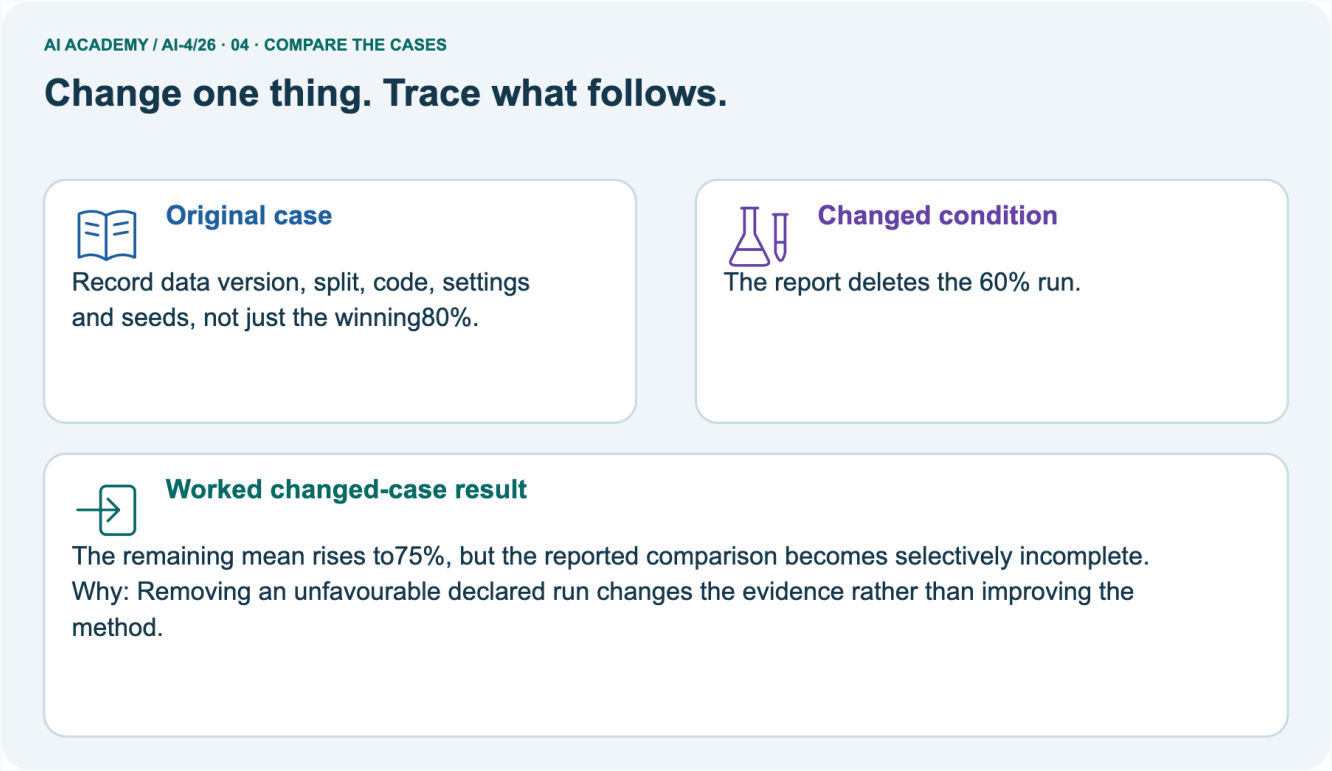


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict deleting the pictured60%run

- Change one thing: The report deletes the **60%** run. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain environment and reporting limits

- **Seed**s help reproduction but do not necessarily remove hardware or library nondeterminism; report the actual environment.

- Read the labelled picture: What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the incomplete record and versioned-policy question

Experimental records and seeds at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Before running	Declare 11 22 33	Save data and settings
Observed results	70 80 60 percent	Keep all three runs
Summary	Mean 70 percent	Range 20 points

The seed is part of the procedure, not evidence that a split represents the target population. Reproducing leakage perfectly reproduces a mistake. Check group separation and legitimate features before celebrating identical outputs.

A file contains only seed 22 and 80 percent. Name four missing items. Explain the steps or evidence behind your response.

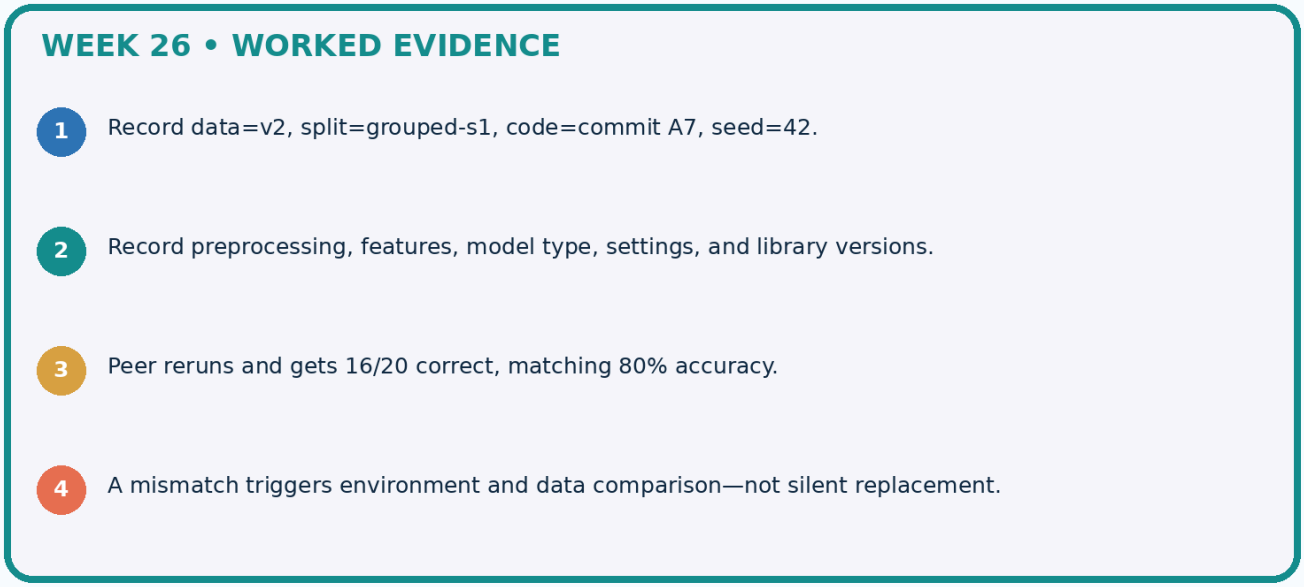
If the data also changes between v1 and v2, can the ranking change be attributed only to weights?

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



STEP 1 Record data=v2, split=grouped-s1, code=commit A7, **seed**=42.

STEP 2 Record pre**process**ing, features, model type, settings, and library **version**s.

STEP 3 Peer reruns and gets 16/20 correct, matching **80%** accuracy.

STEP 4 A mismatch triggers environment and data comparison—not silent replacement.

CHECK Verify the denominator, unit, split, class definition, and **data version** before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

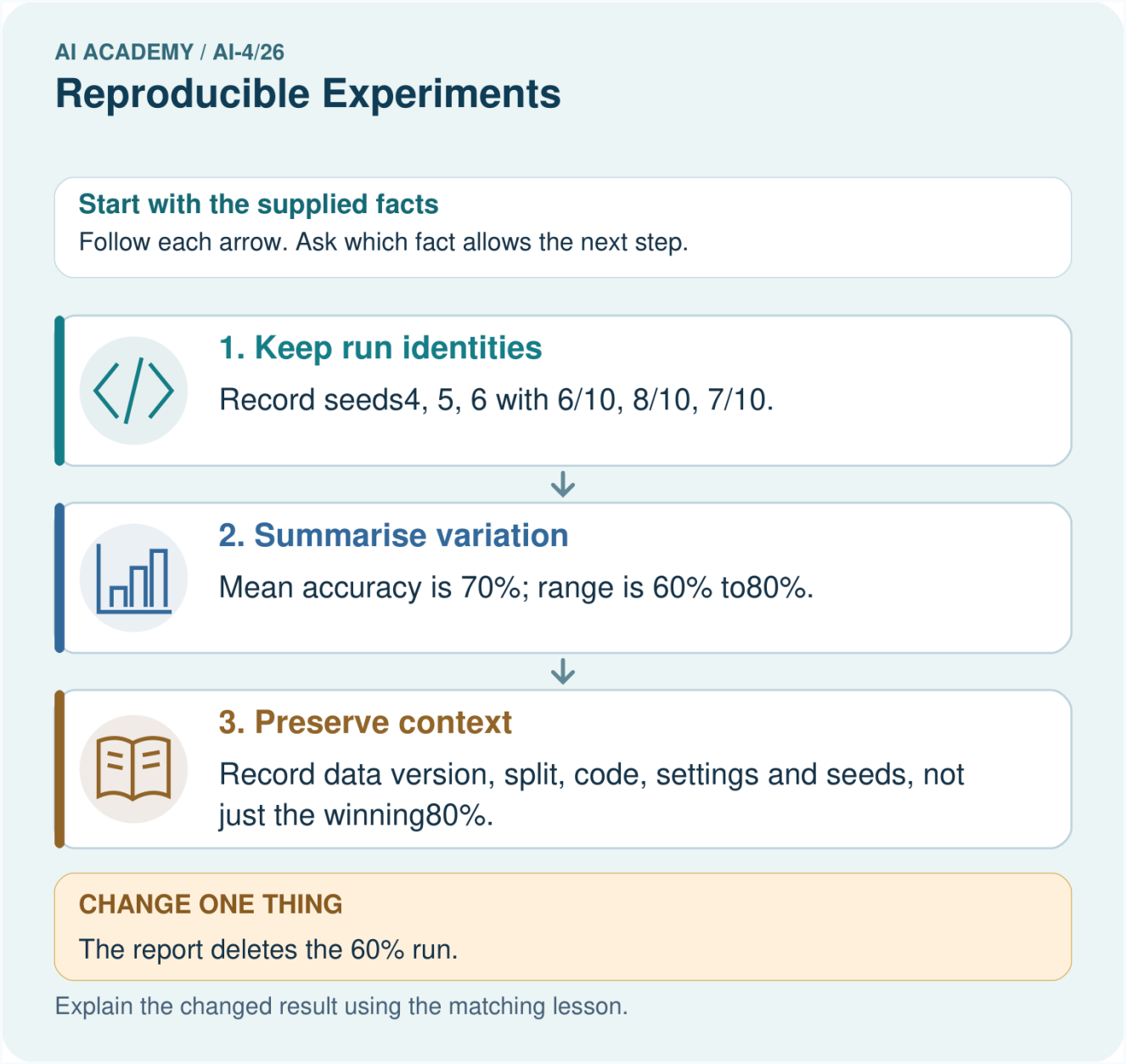
Apply the connected idea: Versioned experiments

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Reproducible Experiments

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a saved seed mistaken for complete reproducibility

> A deterministic wrong result can reproduce exactly.

Diagnose the main claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE Writing down the final accuracy makes an experiment reproducible.

Explain records and limits

- When comparing two methods, use the same declared splits or **seed**s where appropriate and compare paired results. A mean across reused samples does not automatically provide a statistical confidence claim.

  Report the procedure and observed variation plainly.

> **One planned seed fails before prediction. How should it appear in the record?**

Apply the connected idea: Versioned experiments

▼ **Versioned-policy holdout exposure question**

Once **holdout** results guide another change, those cases have become development **evidence**. A later independent evaluation needs new reserved cases. Small **holdout**s still provide limited **evidence**.

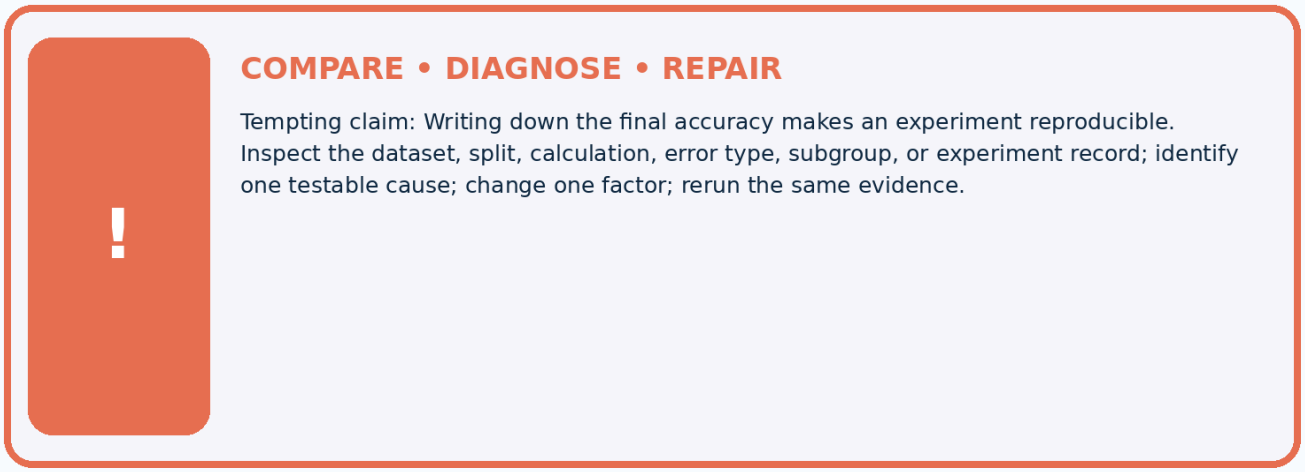
You revise the policy after reading final-case failures. Are those cases still untouched holdout evidence?

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

MAIN LESSON · SECTION 8

Practise improving the record before rerunning

> A version label must resolve to a retained state.

Try supported W2

Use workbook W2 for the connected example “Experimental records and **seed**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Save inputs,outputs,failures and overwritten-version risks

A small record can contain run ID, timestamp, dataset checksum, **training** and validation IDs, model settings, feature **version**, **seed**, **prediction**s and notes. Keep personal information out of exported logs.

Save failed runs with reasons instead of quietly removing them.

Overwriting a file called latest removes the ability to reproduce earlier results unless **version**s are retained elsewhere. Save identifiable **version**s and a run log.

Avoid editing a reserved final test after seeing its **output** to make the chosen **version** look better.

Explain the evidence behind a reproduced score

- Use the worked case for Experimental records and **seed**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain what accompanies the pictured mean

- What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve the independent12-case seed summary

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Additional versioned policy practice

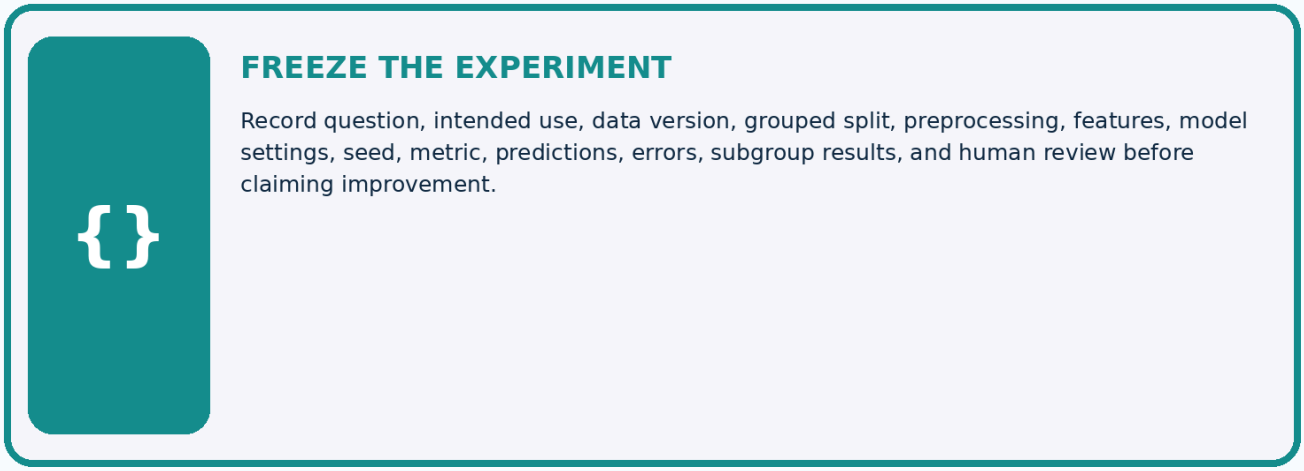
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + **seed**

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Apply the connected idea: Versioned experiments

Practise with less help: Improve a record

Check your reasoning

Practise the connected topic: Versioned experiments

Explain the picture

MAIN LESSON · SECTION 9

Run a complete peer replay and audit an incomplete experiment card

> The lesson distinguishes executable supplied procedure,synthetic saved predictions and missing training evidence.

Complete synthetic saved-output table

Fixed validation ID · Reference · Supplied seed 11 prediction · Supplied seed 22 prediction · Supplied seed 33 prediction

> **Fixed validation ID:** V01
>
> **Reference:** clear
>
> **Supplied seed 11 prediction:** blocked
>
> **Supplied seed 22 prediction:** blocked
>
> **Supplied seed 33 prediction:** blocked

> **Fixed validation ID:** V02
>
> **Reference:** blocked
>
> **Supplied seed 11 prediction:** clear
>
> **Supplied seed 22 prediction:** clear
>
> **Supplied seed 33 prediction:** clear

> **Fixed validation ID:** V03
>
> **Reference:** clear
>
> **Supplied seed 11 prediction:** blocked
>
> **Supplied seed 22 prediction:** blocked

Supplied seed 33 prediction: blocked

Fixed validation ID: V04

Reference: blocked

Supplied seed 11 prediction: clear

Supplied seed 22 prediction: clear

Supplied seed 33 prediction: clear

Fixed validation ID: V05

Reference: clear

Supplied seed 11 prediction: blocked

Supplied seed 22 prediction: clear

Supplied seed 33 prediction: blocked

Fixed validation ID: V06

Reference: blocked

Supplied seed 11 prediction: clear

Supplied seed 22 prediction: blocked

Supplied seed 33 prediction: clear

Fixed validation ID: V07

Reference: clear

Supplied seed 11 prediction: clear

Supplied seed 22 prediction: clear

Supplied seed 33 prediction: blocked

Fixed validation ID: V08

Reference: blocked

Supplied seed 11 prediction: blocked

Supplied seed 22 prediction: blocked

Supplied seed 33 prediction: clear

Fixed validation ID: V09

Reference: clear

Supplied seed 11 prediction: clear

Supplied seed 22 prediction: clear

Supplied seed 33 prediction: clear

Fixed validation ID: V10

Reference: blocked

Supplied seed 11 prediction: blocked

Supplied seed 22 prediction: blocked

Supplied seed 33 prediction: blocked

Fixed validation ID: V11

Reference: clear

Supplied seed 11 prediction: clear

Supplied seed 22 prediction: clear

Supplied seed 33 prediction: clear

Fixed validation ID: V12

Reference: blocked

Supplied seed 11 prediction: blocked

Supplied seed 22 prediction: blocked

Supplied seed 33 prediction: blocked

Fixed validation ID: V13

Reference: clear

Supplied seed 11 prediction: clear

Supplied seed 22 prediction: clear

Supplied seed 33 prediction: clear

Fixed validation ID: V14

Reference: blocked

Supplied seed 11 prediction: blocked

Supplied seed 22 prediction: blocked

Supplied seed 33 prediction: blocked

Fixed validation ID: V15

Reference: clear

Supplied seed 11 prediction: clear

Supplied seed 22 prediction: clear

Supplied seed 33 prediction: clear

Fixed validation ID: V16

Reference: blocked

Supplied seed 11 prediction: blocked

Supplied seed 22 prediction: blocked

Supplied seed 33 prediction: blocked

Fixed validation ID: V17

Reference: clear

Supplied seed 11 prediction: clear

Supplied seed 22 prediction: clear

Supplied seed 33 prediction: clear

Fixed validation ID: V18

Reference: blocked

Supplied seed 11 prediction: blocked

Supplied seed 22 prediction: blocked

Supplied seed 33 prediction: blocked

Fixed validation ID: V19

Reference: clear

Supplied seed 11 prediction: clear

Supplied seed 22 prediction: clear

Supplied seed 33 prediction: clear

Fixed validation ID: V20

Reference: blocked

Supplied seed 11 prediction: blocked

Supplied seed 22 prediction: blocked

Supplied seed 33 prediction: blocked

Complete and incomplete peer cards

Peer reproduction card · Complete supplied record and limitation

Peer reproduction card: Complete scoring replay

Complete supplied record and limitation: Printed program in the mechanism section; input A(fun 4, learning 1), B(fun 2, learning 4); v1 weights(2, 1), v2(1, 2); same loop and input; larger score ranks first; declared alphabetical name tie rule if needed. No stochastic step or accuracy target.

Peer reproduction card: Record your actual environment

Complete supplied record and limitation: Write actual Python version, code copy/version, input/weight versions and exact outputs. Do not claim the lesson supplies your local environment.

Peer reproduction card: Expected deterministic outputs

Complete supplied record and limitation: v1A9/B8; v2A6/B10. A different file/input/weight version must be compared before attributing a mismatch.

Peer reproduction card: Intentionally incomplete classifier card

Complete supplied record and limitation: Seed22, accuracy 80%. Missing actual data/label state, split membership, code, preprocessing/features, model/settings, environment, metric/failure policy and per-case outputs. Do not invent them.

Peer reproduction card: Saved-output replay fixture

Complete supplied record and limitation: The twenty-row Vtable is a synthetic record matching the supplied 14/16/12 counts; it allows auditing the arithmetic, not rerunning an unknown classifier.

Normal,same-seed,missing-record and selective-report probes

Complete reproducibility probe	Expected bounded check
Normal same input/loop/weights in the deterministic scorer	v1[9, 8], v2[6, 10]; a seed is not used.
Boundary same seed but different label file	Not the same experiment; preserve both identifiable data/label states and compare them.
Missing only seed 22 and80%available	Identify absent records; the training run cannot be independently recreated from this alone.
Changed report hides the lower declared run	Keep the original complete run set and disclose the changed summary instead of claiming an improved method.

Exchange,rerun,audit and explain mismatches

1. Reproduction Swap: first exchange the complete deterministic scoring card, run the printed versioned program in a fresh namespace, and compare exact outputs and actual environment records. Then exchange the intentionally incomplete seed-only classifier card and list the missing evidence without guessing it.

   Audit the complete synthetic Vtable to recover the three correct counts and full mean/range; label this a saved-output arithmetic replay. Preserve all declared runs and any failures. Keep the selected seed 1/2/3on12 cases for an uncoached first summary attempt.

Optional complete saved-output summary program

```
# Complete synthetic saved-output record for arithmetic replay, not model training.
references = ['clear' if i % 2 else 'blocked' for i in range(1, 21)]
predictions = {}
for seed, mistakes in [(11, 6), (22, 4), (33, 8)]:
    predictions[seed] = [('blocked' if y == 'clear' else 'clear') if i < mistakes else y
                         for i, y in enumerate(references)]
accuracies = []
for seed in [11, 22, 33]:
    correct = sum(y == p for y, p in zip(references, predictions[seed]))
    accuracies.append(correct / len(references))
    print(seed, correct, len(references))
print('mean', round(sum(accuracies) / len(accuracies), 6))
print('range percentage points', round(100 * (max(accuracies) - min(accuracies)), 6))
# Seeds label supplied run records here; no classifier or random process is rerun.
```

Retain evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Reproduction Swap: exchange incomplete experiment cards, identify missing fields, then reproduce and explain any mismatch.

1. Write the hypothesis and expected **evidence** before running.

2. Reproduction Swap: exchange incomplete experiment cards, identify missing fields, then reproduce and explain any mismatch.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: summarize three fresh declared runs

> Keep the numerator,denominator and range unit visible.

Complete independent results and scope

Declared **seed**s 1, 2, 3 produce 9/12, 8/12, 10/12 correct. Compute the mean accuracy and range.

Use all three declared runs with the same 12-case denominator. Report mean run accuracy and the max-minus-min range in percentage points. The task supplies aggregate outcomes only; it does not supply per-case predictions or an executable classifier, so do not claim that the arithmetic reconstructs training. Retain every run and state the limited evidence.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply independently

Week 26 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Two runs use different random splits and different feature sets. Their scores differ. What must be recorded, and why is the cause unclear?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Give every Plant Classifier experiment an ID and immutable record linking dataset, split, code, settings, **prediction**s, and metrics.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Two teams use the same **random seed** but different label files. Their results differ. List the records needed to reproduce the experiment.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep an experiment identity that resolves to actual saved evidence

> Do not let a label such as latest replace version history.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Give every Plant Classifier experiment an ID and immutable record linking dataset, split, code, settings, **prediction**s, and metrics.

Give every project run an ID linked to retained data/label versions, source split IDs, code/dependency versions, features/preprocessing, fitted state/settings, seeds or explicit no-random-step status, environment, metric policy and raw outputs. Record planned versus actually run attempts and failures.

Reproducing an output does not establish its truth, benefit or independence; retain development exposure and the untouched final-evaluation boundary.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric **version** does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

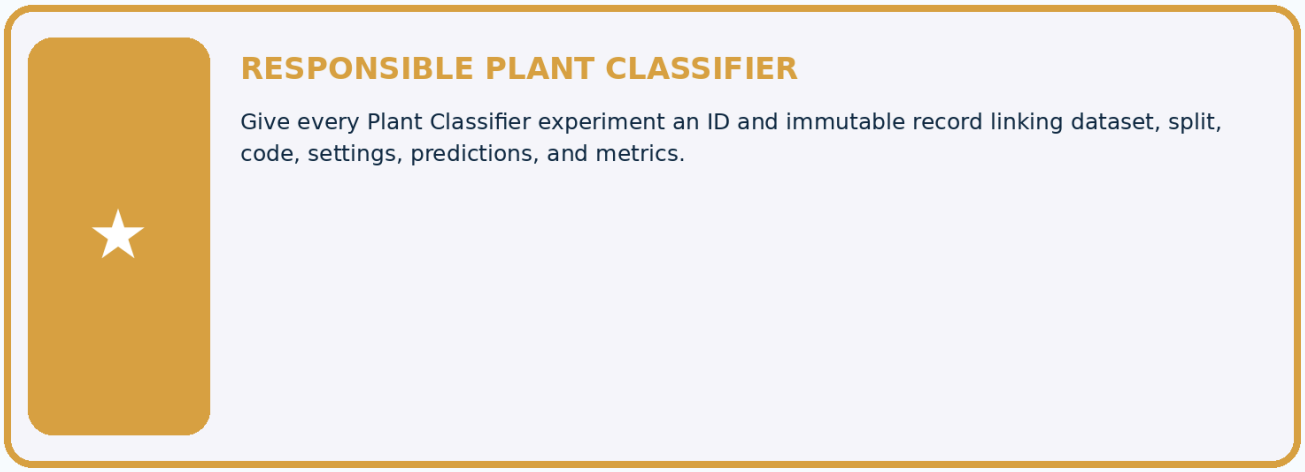
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

Finish with what repeated and what remains unidentified

> Explain why the best score and a seed are not a complete experiment.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Record enough detail to reproduce a run

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
reproducibility	ability to recreate a result	
data version	identified dataset state	
random seed	value controlling repeatable randomness	
experiment record	complete inputs, settings, and outcomes	

Closer-explanation terms

Word	Meaning
Seed	An initial value controlling a pseudo-random process
Run record	A saved account of inputs, settings and outputs
Reproducibility	Ability to repeat a specified procedure

Word	Meaning
Cherry-picking	Reporting favorable results while hiding relevant others

▼ Extension vocabulary

Term	Meaning
Version	An identified state of code or rules
Configuration	Settings used for a run
Development set	Cases used while choosing or improving a method
Holdout	Cases reserved for an evaluation after choices are fixed

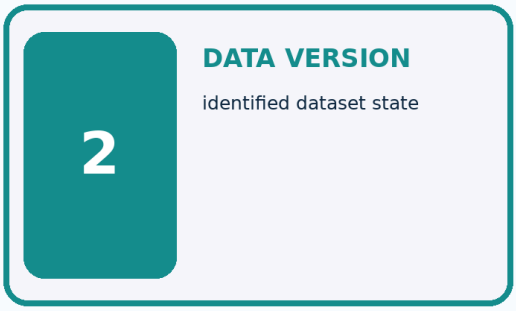
▼ Earlier wording, alternatives and picture transcripts

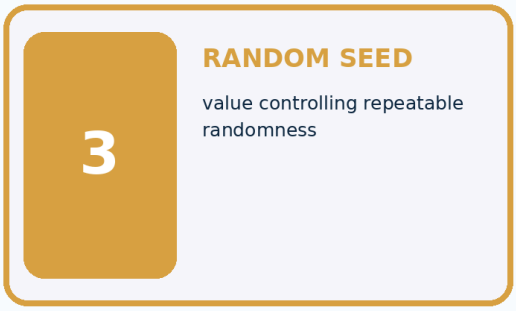
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Writing down the final accuracy makes an experiment reproducible.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Terms for Versioned experiments

▼ Extra challenge · Build a stronger explanation

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **reproducibility**, **data version**, **random seed**, **experiment record**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Writing down the final accuracy makes an experiment reproducible.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

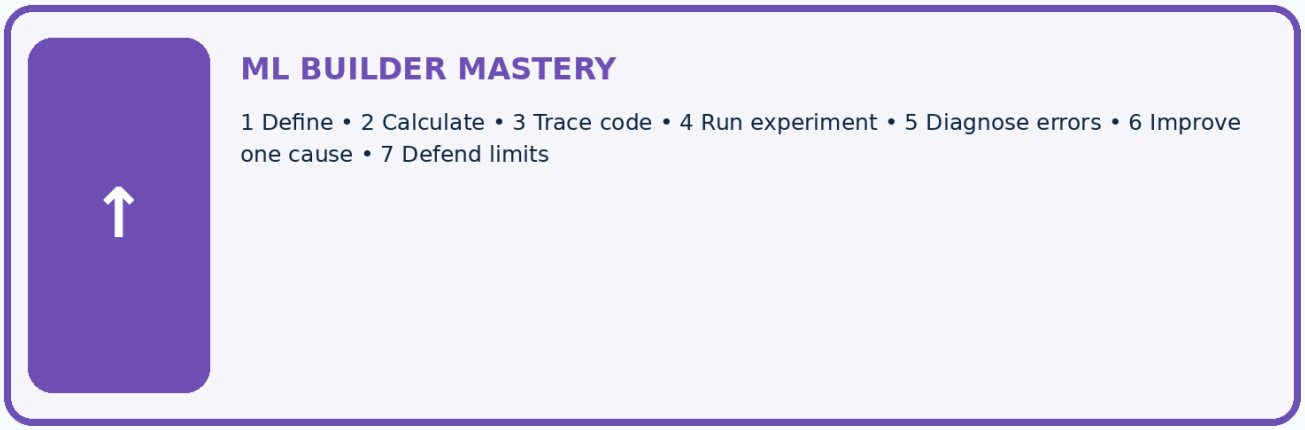
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh seed1/2/3twelve-case summary task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or **experiment record**.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

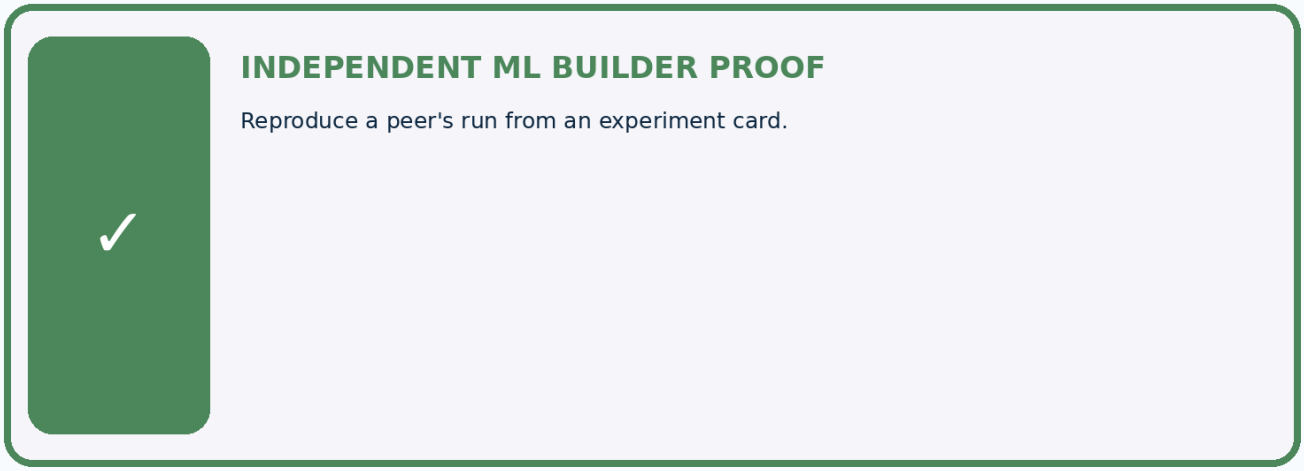
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Reproduce a peer's run from an experiment card.

NEXT BRIDGE Carry the experiment **evidence** into: Over**fitting**.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

A report says accuracy85% without data split, code **version**, **seed** or pre**process**ing.

Specify **reproducibility** records.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Record data identity, group/split IDs, code/dependency **version**s, pre**process**ing, features, settings, **seed**s where relevant, metric definition and raw **prediction**s. **Seed** alone is insufficient.

Evidence or stage	Value or result
Data	Identity and split/group IDs
Code	Version and dependencies
Procedure	Features, preprocessing and settings
Evidence	Seeds, metric definitions and raw predictions

▼ Earlier wording, alternatives and picture transcripts

Week 26 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Make an experiment reproducible

A report says accuracy85% without data split, code **version**, **seed** or pre**process**ing.

Specify **reproducibility** records.

Record data identity, group/split IDs, code/dependency **version**s, pre**process**ing, features, settings, **seed**s where relevant, metric definition and raw **prediction**s. **Seed** alone is insufficient.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Record enough detail to reproduce a run	Explain the deciding step, show a trace or test, and state one limit.
Distinguish reproducibility from validity	Explain the deciding step, show a trace or test, and state one limit.
Report all planned seeds instead of selecting the best	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Record variation across declared **seed**s instead of selecting the prettiest run.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Record variation across declared **seed**s instead of selecting the prettiest run.

- Which labelled step matches this detail? Record **data version**, split, code, settings and **seed**s, not just the winning**80%**.
- 1. Preserve context
- 2. Keep run identities
- 3. Summarise variation
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Preserve context Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. Read the picture in order.

Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The report deletes the **60%** run. Which conclusion is supported by the supplied facts?

- 1. The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.

Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Repair and keep the record

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised **version**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

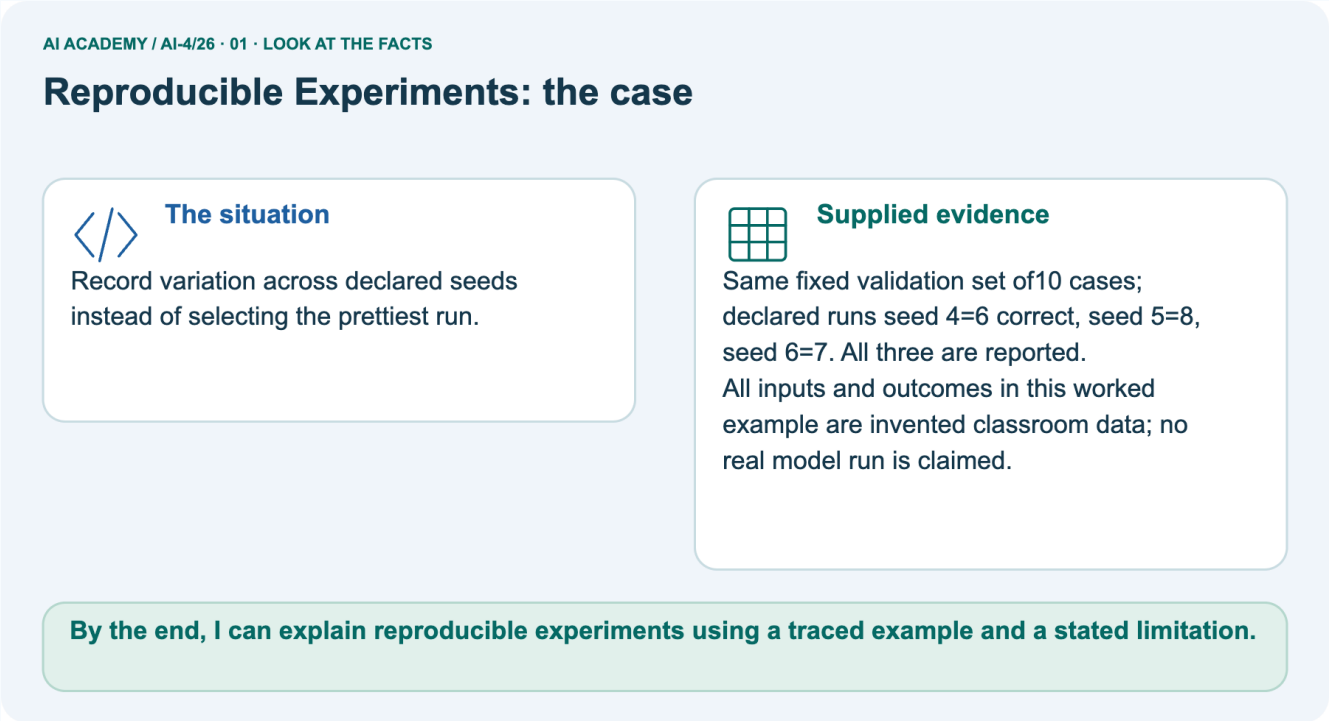
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Reproducible Experiments: the case



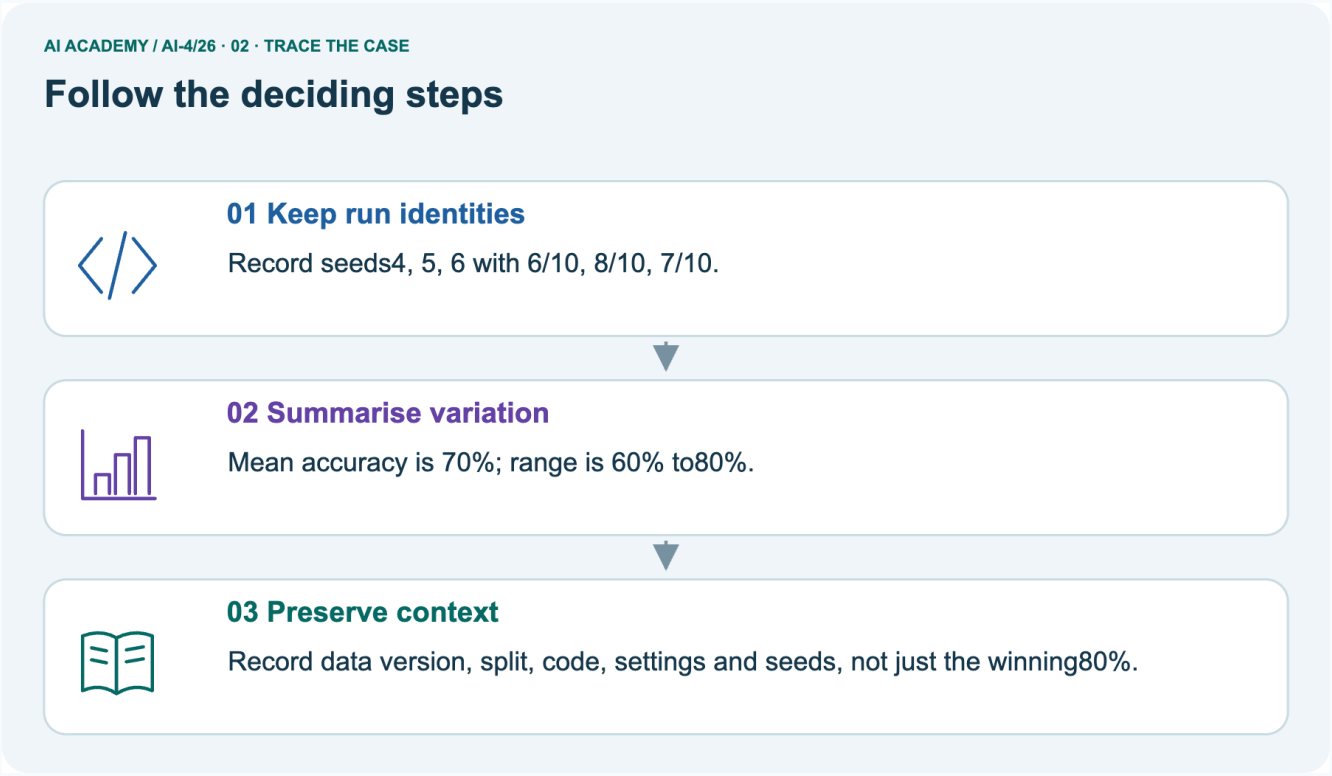
Follow the labelled picture with the matching explanation. Open the vector diagram

Record variation across declared **seed**s instead of selecting the prettiest run. Supplied facts: Same fixed validation set of10 cases; declared runs **seed** 4=6 correct, **seed** 5=8, **seed** 6=7. All three are reported.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep run identities: Record **seed**s4, 5, 6 with 6/10, 8/10, 7/10. → Summarise variation: Mean accuracy is **70%**; range is **60%** to**80%**. → Preserve context: Record **data version**, split, code, settings and **seed**s, not just the winning**80%**.

Point to each numbered stage and explain which recorded **input** it uses.

03. All declared seeds

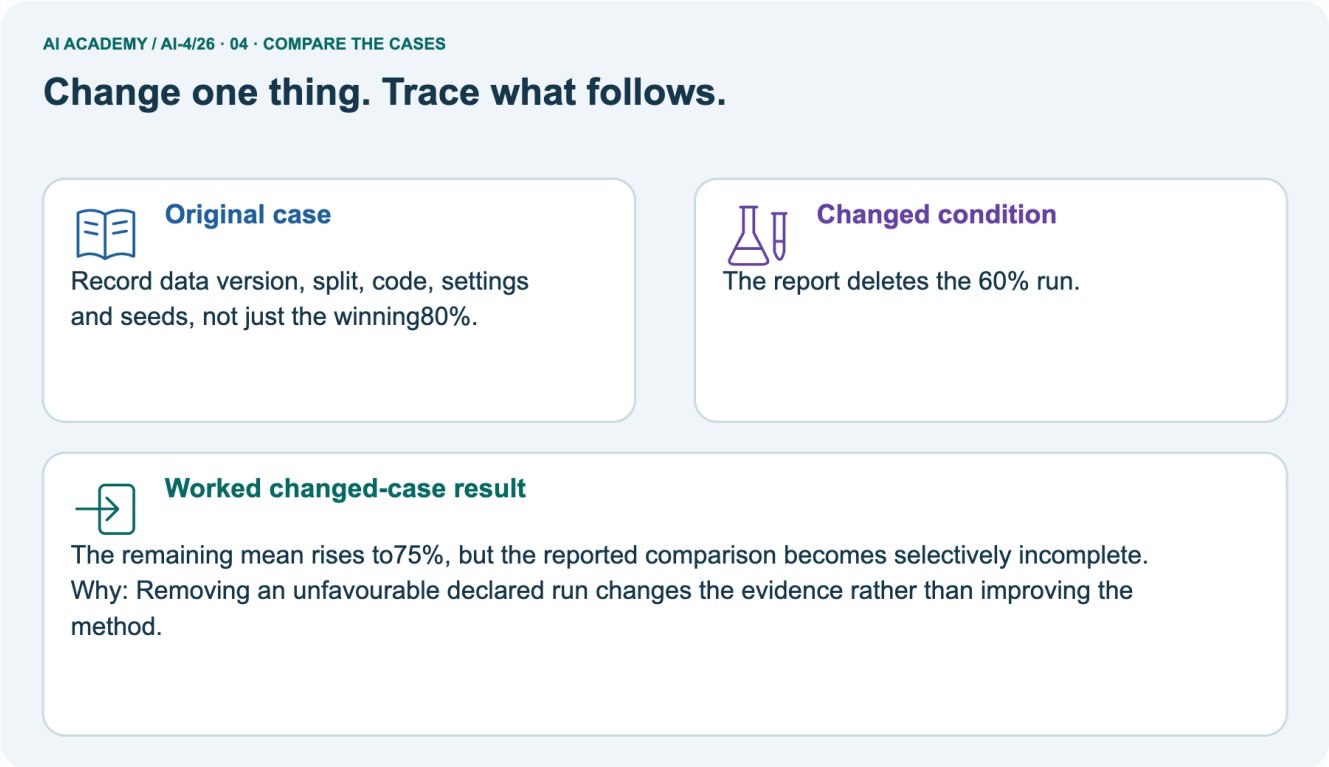


Follow the labelled picture with the matching explanation. Open the vector diagram

All declared **seed**s. Units: accuracy percent. **seed** 4: 60. **seed** 5: 80. **seed** 6: 70.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



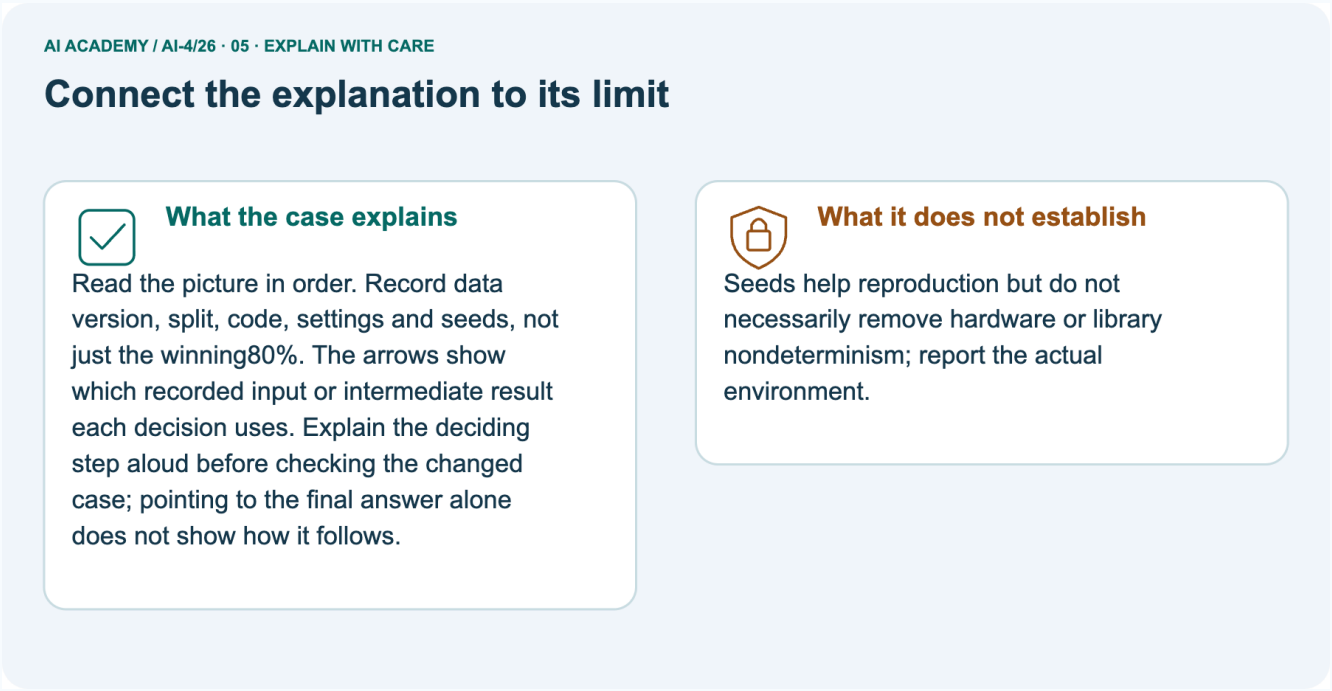
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. Changed condition: The report deletes the **60%** run.

Expected result: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete. Reason: Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

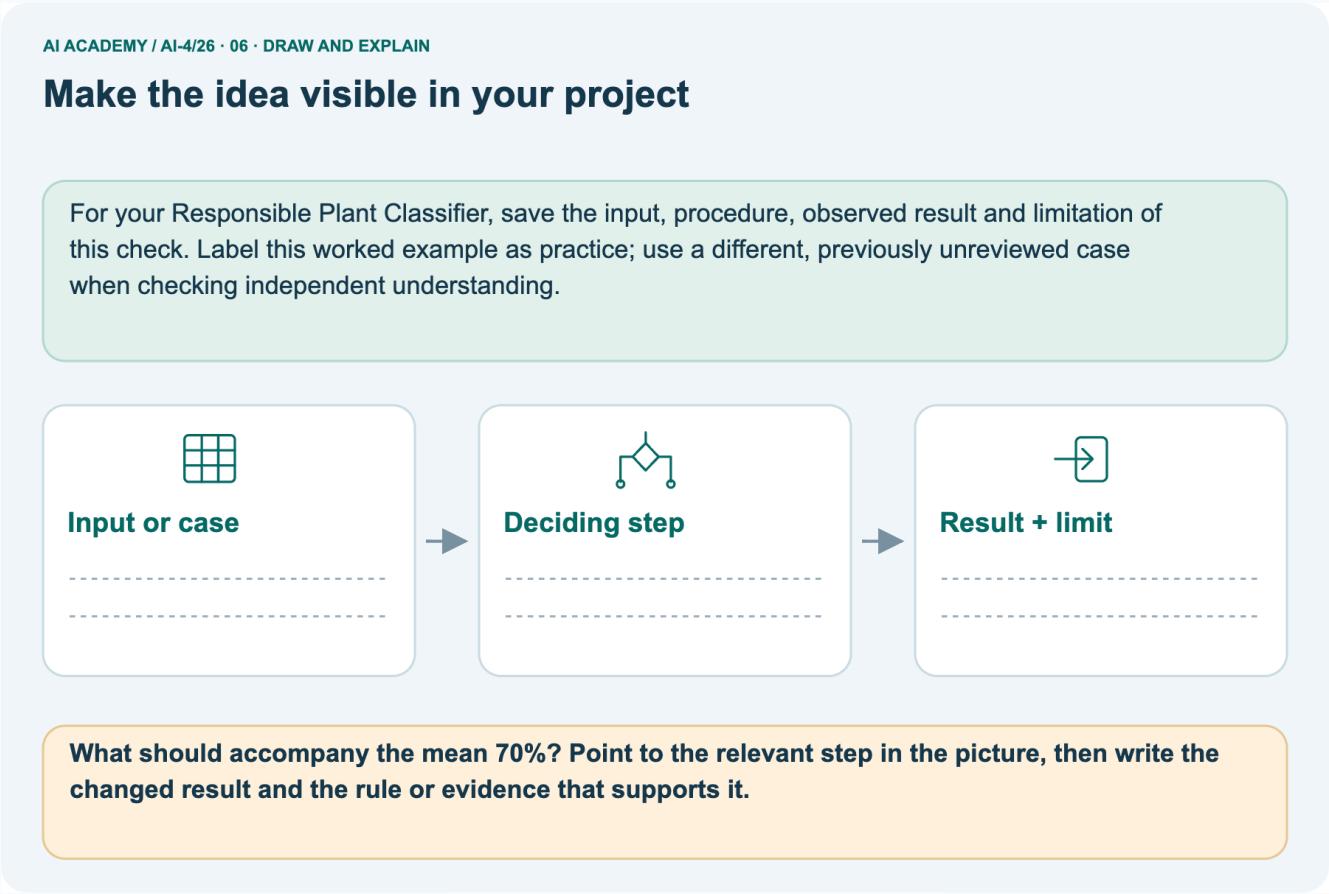
Read the picture in order. Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Seed**s help reproduction but do not necessarily remove hardware or library nondeterminism; report the actual environment.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Earlier wording, alternatives and picture transcripts

01. Reproducible Experiments: the case

02. Follow the deciding steps

03. All declared seeds

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.